

ELEDIA

Branch Office User Upload

User manual for administrators
and branch managers

Plugin local_branchupload
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eLeDia GmbH
Wilhelmsaue 37 · 10713 Berlin · Germany
eledia.de · info@eledia.de

Branch Office User Upload — User Manual

local_branchupload · version 1.4.0

eLeDia GmbH · Christopher Reimann

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Overview

The Moodle plugin "**Branch Office User Upload**" (`local_branchupload`) lets staff in your branch offices upload member lists as CSV files into Moodle — **without** giving them administrator rights. Every user that is uploaded is automatically

- created as a Moodle user (or updated if they already exist),
- assigned to the correct branch office (as a cohort),
- given an organisational unit (stored in a custom profile field),
- e-mailed their login credentials.

The plugin was originally built for **Landkreis Ravensburg**, a German county-level federation of municipalities, but is generic enough for any organisation with a *branch-office* structure — a school district, a corporate group with subsidiaries, a training company with regional locations, a multi-tenant SaaS deployment, ...

Security model at a glance

Each branch may **only manage users that belong to it**:

- The uploader's own branch is determined by a **custom profile field** on their Moodle account (default name: "*Behörde*").
- The uploader can only create users for **their own** branch.
- Existing users that belong to a **different** branch cannot be modified.
- *Smuggling* a foreign branch's cohort through the `Cohorts` column is also blocked.

- **Site administrators** are exempt from these restrictions and can upload for any branch.

What's new in 1.4.0?

- **English canonical keys** (`email`, `branch`, `orgunit`, `lastname`, `firstname`, `remove`, `cohorts`, `oldemail`) are used as the internal identifiers. As an administrator the only thing you see is that the plugin-configuration keys are now in English. The default *CSV column headers* you have to put in your file remain locale-correct.
 - **Language-dependent defaults.** The default column headers are now resolved at runtime from the configured *site language* (`$CFG->lang`). A German site sees `Behörde / Organisationseinheit / Name / Vorname / Löschen / Kohorten / Alte_Email`; an English site sees `Branch / OrgUnit / LastName / FirstName / Remove / Cohorts / OldEmail` — automatically, with no code changes.
 - **Automatic migration.** If you had custom CSV-header overrides configured in 1.3.0, the new `db/upgrade.php` migrates those values from the old German config keys (`col_behoerde`, `col_orgeinheit`, ...) to the new English ones (`col_branch`, `col_orgunit`, ...) as part of the standard Moodle upgrade. No manual action is required.
-

For administrators: setting up the plugin

Important. The following steps must be performed **once** by a Moodle administrator before branch managers can use the upload page.

Installation

1. Copy the `branchupload` folder into your Moodle installation's `local/` directory. On Moodle 5.x this lives under the new `public/` document root:

```
{moodle_root}/public/local/branchupload/
```

On Moodle 4.x:

```
{moodle_root}/local/branchupload/
```

2. Sign in as a site administrator and navigate to **Site administration** → **Notifications**. Moodle will detect the new plugin and run its installation. Alternatively, trigger the install from the command line:

```
php admin/cli/upgrade.php --non-interactive
```

3. After installation the plugin's configuration page appears. You can always come back to it later at **Site administration** → **Plugins** → **Local plugins** → **Branch office user upload**.

Create the two custom profile fields

The plugin requires **two custom user profile fields**. Create them *before* you configure the plugin.

Navigate to: Site administration → Users → Accounts → User profile fields

Profile field 1 — Branch office

This field identifies which branch a user (and an uploader) belongs to.

Setting	Recommended value
Field type	Text (or Menu of choices if the branch list is fixed)
Short name	branchoffice
Name	Behörde (or your localised equivalent, e.g. "Branch")
Required	No (the plugin manages it automatically)
Visible to	Everyone

Tip for a Menu of choices. Add one branch identifier per line, e.g.

```
GmndAchbrg
GmndWangen
GmndIsny
StadtRV
```

The values must match the cohort `idnumber` *exactly* (see next section).

Profile field 2 — Organisational unit

This field stores the department or organisational unit from the CSV file.

Setting	Recommended value
Field type	Text
Short name	orgunit
Name	Organisationseinheit (or "Organisational unit")
Required	No
Visible to	Everyone

Create one cohort per branch

Each branch office is represented by a **cohort**. The cohort's **idnumber** must match the value used in the **Branch** (German: **Behörde**) column of the CSV.

Navigate to: Site administration → Users → Accounts → Cohorts

For each branch, create a cohort:

Setting	Example value
Name	Achberg Municipality
Cohort ID	GmndAchbrg
Context	System

Repeat for every branch, e.g.:

Name	Cohort ID
Achberg Municipality	GmndAchbrg
Wangen Municipality	GmndWangen
Isny Municipality	GmndIsny
Ravensburg City	StadtRV

Alternative. In the plugin settings you can enable "*Auto-create cohorts*". Missing cohorts referenced in a CSV will then be created on the fly. We recommend this **only during initial roll-out**, not in steady-state production.

Configure the plugin settings

Navigate to: Site administration → Plugins → Local plugins → Branch office user upload

Setting	Description	Recommendation
Branch profile field	Pick the " <i>Behörde</i> " profile field you created above.	branchoffice
Organisational unit profile field	Pick the " <i>Organisationseinheit</i> " profile field.	orgunit
Auto-create cohorts	If on, cohorts referenced in the CSV but missing from Moodle are created automatically.	In production: off
Delete action	What <code>Remove=1</code> does. "Suspend" disables the account (reversible). "Delete" removes it permanently.	Suspend (safer default)
Maximum users per upload	Per-CSV row cap. <code>0</code> disables the limit.	500

Optional — customise the CSV column headers

The "CSV *column headers*" settings section lets you rename every column header to match the files your branch offices already produce. The plugin's internal identifiers (English, ASCII-only) never change between releases — what changes is just the *string* that has to appear in the first row of the CSV.

Configuration key	Re-quired?	Default (German site)	Default (English site)
local_branchupload/col_email	yes	Email	Email
local_branchupload/col_branch	yes	Behörde	Branch
local_branchupload/col_orgunit	yes	Organisationseinheit	OrgUnit
local_branchupload/col_lastname	yes	Name	LastName
local_branchupload/col_firstname	yes	Vorname	FirstName
local_branchupload/col_remove	no	Löschen	Remove
local_branchupload/col_cohorts	no	Kohorten	Cohorts
local_branchupload/col_oldemail	no	Alte_Email	OldEmail

Column-header matching properties:

- **Case-insensitive** (Email, email, EMAIL all match).
- **Trim-insensitive** (leading/trailing whitespace is ignored).
- An **empty value** in the settings restores the site-language default.

Renamed headers are propagated automatically — no code change required — to:

- the *hint text* above the file picker on the upload form,
- the *downloadable example CSV* on the upload page,
- the column titles in the *preview* and *results* tables,
- the *validation error message* when a required column is missing.

Upgrade note (1.3.0 → 1.4.0). If you had configured custom headers in 1.3.0 under the old German config keys (col_behoerde, col_orgeinheit, col_name, col_vorname, col_loeschen, col_kohorten, col_alte_email), db/upgrade.php migrates those values to the new English keys on upgrade. No manual action required.

Grant the upload capability

The plugin ships a single capability: **local/branchupload:upload** .

Give it to the people who should be allowed to upload. There are two equivalent ways:

Option A — via a role (recommended)

1. **Site administration** → **Users** → **Permissions** → **Define roles**.
2. Create a new role (e.g. "Branch manager") or edit an existing one.
3. Under **Filter permissions**, search for `branchupload`.
4. Set `local/branchupload:upload` to **Allow**.
5. Assign that role to the relevant users *at system level*: **Site administration** → **Users** → **Permissions** → **Assign system roles**.

Option B — directly per user

1. **Site administration** → **Users** → **Permissions** → **Assign system roles**.
2. Pick the role that contains the upload capability.
3. Add the users.

Set each uploader's own branch value

Critical. Every user who should be able to upload must themselves carry a value in the "Behörde" profile field. That value determines which branch they may upload users for.

1. Open the uploader's profile.
2. Edit the profile and set the "Behörde" field to the corresponding cohort `idnumber` (e.g. `GmndAchbrg`).
3. **Without this value the uploader cannot perform any upload** and will see an explanatory error message.

Example. Ms Müller is the responsible person for Achberg Municipality. Her profile has *Behörde* = *GmndAchbrg*. She can therefore only upload CSV files where every row's *Branch* column equals *GmndAchbrg*.

For branch managers: uploading users

You can hand this section directly to your branch-office staff.

Prepare your CSV file

The user list is uploaded as a **CSV file** (a plain text file with one user per line, columns separated by a delimiter). You can produce the file in Excel, LibreOffice Calc, Numbers or any text editor.

Important rules

- **Delimiter:** semicolon (;) is the default. You can pick a different delimiter during upload.
- **Encoding:** UTF-8 — so umlauts and accented characters (ä, ö, ü, é, ñ, ...) survive correctly.
- **First row = column headers.** The first row of the file contains the column names (see below).

Excel tip. When saving, choose "CSV UTF-8 (Comma delimited)". On a German Excel installation the actual separator used is the semicolon.

Required columns

Every CSV file **must** contain these columns (default English headers for an English-language Moodle site):

Column	Description	Example
Email	The user's e-mail address. Also used as the Moodle username.	max.mustermann@example.de
Branch	Branch identifier — must match a cohort <code>idnumber</code> .	GmndAchbrg
OrgUnit	The user's department or organisational unit.	Bauverwaltung
Last-Name	Family name.	Mustermann
First-Name	Given name.	Max

Note for German Moodle sites. The default headers on a German site are Email / Behörde / Organisationseinheit / Name / Vorname instead. The example CSV you download from the upload page always uses the headers configured for your site, so you don't need to guess.

Optional columns

These columns *may* appear in addition to the required ones:

Column	Description	Accepted values
Re-move	Mark a user for removal. Empty or missing = no removal.	<code>1</code> , <code>ja</code> , <code>yes</code> or <code>true</code> .
Cohorts	Additional cohort assignments (on top of the user's branch cohort). Multiple cohorts are separated by a pipe character (<code> </code>).	<code>`</code> .
OldE-mail	The user's previous e-mail when renaming them. The plugin finds the existing user by the old e-mail and updates them with the new e-mail (from the <code>Email</code> column) and the new username.	<code>previous.address@example.de</code>

Note about cohorts. You cannot use the `Cohorts` column to add a user to a *branch* cohort. Branch cohorts (e.g. `GmndAchbrg`) may only be assigned via the `Branch` column. This prevents a branch manager from sneaking their users into another branch via the cohorts column. Site administrators are exempt from this restriction.

What a complete CSV file looks like

```
Email;Branch;OrgUnit;LastName;FirstName;Remove;Cohorts;OldEmail
max.mustermann@example.de;GmndAchbrg;Bauverwaltung;Mustermann;Max;;TrainingA;
erika.neu@example.de;GmndAchbrg;Finanzen;Musterfrau;Erika;;TrainingA|TrainingB;erika.musterfrau@
hans.beispiel@example.de;GmndAchbrg;Ordnungsamt;Beispiel;Hans;1;;
```

Row-by-row explanation:

- **Row 1.** Max Mustermann is created as a new user, joined to the `GmndAchbrg` cohort (his branch) and additionally added to `TrainingA`.
- **Row 2.** Erika Musterfrau changed her e-mail address: she is located by her previous address (`erika.musterfrau@example.de`), updated to her new address (`erika.neu@example.de`) — including a new username — and added to two extra cohorts.
- **Row 3.** Hans Beispiel is *suspended* (or *deleted*, depending on the administrator's chosen action) because `Remove=1`.

Download the example CSV

The upload page provides a "Download example CSV" link. The example file is generated **on the fly** from the column headers configured for your site — so if your administrator renamed the columns to e.g. `EmailAddress;Site;Department;...`, the downloaded example uses exactly those header names.

Step-by-step: performing an upload

Step 1 — Open the upload page

Sign in to Moodle. You'll find the "*Upload branch users*" link in any of these places:

- the **left navigation drawer**,
- your **profile** page,
- if you're an administrator: **Site administration** → **Plugins** → **Local plugins** → **Upload branch users**.

Step 2 — Upload the CSV file

1. Click "**Choose a file**" and pick your CSV (or drag-and-drop into the upload area).
2. **CSV delimiter**: pick the delimiter your file uses (default: semicolon).
3. **Encoding**: leave at UTF-8 unless your file uses a different encoding.
4. Click "**Upload CSV**".

Step 3 — Review the preview

After upload you see a **preview table** with every row of your file:

- **Above** the table your own branch is shown (e.g. "*Your branch: GmndAchbrg*").
- Each row has a **status badge**:
 - **Green — Will be created**: a new user will be created.
 - **Blue — Will be updated**: the user already exists and will be updated.
 - **Yellow — Will be suspended**: the user will be deactivated (*Remove=1*).
 - **Red — Error**: the row has a problem and will be skipped. The reason is shown next to the badge.
- A **summary line** above the table shows the total row count, plus how many rows are valid and how many have errors.

Review the preview carefully! Only rows marked valid will be processed — rows with errors are silently skipped.

Step 4 — Confirm the upload

- If everything looks correct, click "**Confirm upload**".
- If you spot a problem, click "**Cancel**", fix your CSV and start over.

Step 5 — Review the results

After processing, a **results overview** shows the counters *Users created*, *Users updated*, *Users suspended*, *Users deleted*, *Rows skipped* and *Errors* — followed by a per-row table with the outcome of each row.

What about credentials? Newly created users receive their login details **automatically by e-mail**. Delivery happens through Moodle's cron and may take a few minutes. New users are prompted to change their password on first login.

Frequently asked questions

"I can't upload any users — I get an error."

Likely causes:

1. **No capability.** Your administrator must grant you `local/branchupload:upload`.
2. **No branch value.** The `Behörde` field on your own profile is empty. Ask your administrator to set it.
3. **Plugin not configured.** The administrator must select the two profile fields in the plugin settings.

"Some rows show as errors."

Look at the error message displayed for the row. Common reasons:

- **Invalid e-mail** — missing `@`, typo, ...
- **Branch mismatch** — the row's `Branch` column does not equal your own branch.
- **Cohort does not exist** — the cohort `idnumber` is unknown and auto-creation is disabled.
- **Required field empty** — `LastName`, `FirstName`, `Email`, `Branch` or `OrgUnit` is missing.

"What happens when I upload a file containing an existing user?"

The user is **updated**: first name, last name, organisational unit and cohort memberships are set to the values in the CSV. The **password** is *not* changed.

"How can I change a user's e-mail address?"

Use the optional `OldEmail` column: put the *new* e-mail in the `Email` column and the *current* (about-to-be-replaced) e-mail in the `OldEmail` column. The plugin finds the user by the old e-mail and updates them with the new e-mail and username.

Example:

```
Email;Branch;OrgUnit;LastName;FirstName;OldEmail
new.address@example.de;GmndAchbrg;Bauverwaltung;Mustermann;Max;previous.address@example.de
```

"Can I edit users belonging to another branch?"

No. You can only edit users that belong to your own branch. Any attempt to update users from another branch is rejected with an explicit error message.

"What does `Remove=1` actually do — is the user deleted entirely?"

It depends on your administrator's chosen action:

- **Suspend** (the documented default): the account is disabled. The user can no longer sign in, but all their data is preserved and the account can be re-enabled later.
- **Delete**: the account is removed permanently. This cannot be undone.

"Umlauts in my CSV file appear garbled."

Make sure the file is saved as **UTF-8**:

- **Excel:** save as "CSV UTF-8 (Comma delimited)".
- **LibreOffice:** choose UTF-8 in the export dialog.
- **Upload form:** pick the matching encoding (default UTF-8).

"When do users actually receive their credentials?"

New users receive an e-mail with their username and password as soon as the next **Moodle cron** run completes — usually within a few minutes of the upload. If credential mails do not arrive, contact your administrator.

"Can we use English column headers instead of German ones?"

Yes — in two equivalent ways:

1. **Per-header overrides.** In the plugin settings under "CSV column headers", rename every column header individually to any string you like.
 2. **Site language.** If your Moodle site language (`$CFG->lang`) is set to English, the defaults are automatically `Branch / OrgUnit / LastName / FirstName / Remove / Cohorts / OldEmail` — no per-setting overrides required.
-

Error messages reference

Error message	What it means	What to do
"Missing required columns: ..."	The CSV file is missing one or more required column headers.	Check the first row of the file. Required columns are the headers configured in the plugin settings — by default <code>Email</code> , <code>Branch</code> , <code>OrgUnit</code> , <code>LastName</code> , <code>FirstName</code> (English site) or <code>Email</code> , <code>Behörde</code> , <code>Organisationseinheit</code> , <code>Name</code> , <code>Vorname</code> (German site).
"Branch mismatch"	The row's <code>Branch</code> value differs from your own branch.	Fix the value in the <code>Branch</code> column, or have the upload performed by the relevant branch.
"Cohort ... does not exist"	The cohort <code>idnumber</code> is unknown and auto-creation is disabled.	Ask your administrator to create the cohort, or to enable auto-creation.
"Invalid email address"	The e-mail format is malformed.	Check the address for typos.
"Email is required"	The <code>Email</code> column is empty in this row.	Provide an e-mail address.
"Cannot update user — different branch"	The user belongs to a different branch than yours.	Only the user's own branch can update them.
"The new email address ... is already used by another user"	The target e-mail is taken by another active user.	Double-check that the new address is correct.
"Invalid previous email address"	The format of the <code>OldEmail</code> value is malformed.	Check the old e-mail address.
"Cohort ... represents a branch office and cannot be assigned via the cohorts column"	A branch cohort was specified in the <code>Cohorts</code> column.	Branch cohorts may only be set via the <code>Branch</code> column.
"Profile field not configured"	The administrator has not finished configuring the plugin.	Contact your administrator.

Error message	What it means	What to do
"Your user account does not have a branch value set"	The Behörde field on your own profile is empty.	Ask your administrator to set the value.
"The CSV contains ... rows, but the maximum is ..."	The file exceeds the per-upload row cap.	Split the file into smaller chunks.
"Error parsing CSV file"	The file could not be parsed.	Check the file format, encoding and delimiter.

Technical notes

System requirements

- **Moodle** 4.5 (build 2024100700) or newer — including Moodle 5.x.
- **PHP** 8.1, 8.2, 8.3 or 8.4.
- **Databases** MariaDB / MySQL and PostgreSQL (covered by the CI matrix).

Data handling

- Uploaded CSV files are **not persisted**. They are processed in memory by Moodle's `csv_import_reader` and discarded after processing.
- New users get auto-generated passwords sent by e-mail. The passwords are stored hashed in Moodle's database — never in plain text.
- Users are prompted to change their password on first login.

Plugin file structure

local/branchupload/	
├─ version.php	– plugin version (1.4.0) and requirements
├─ index.php	– upload page (three-step flow)
├─ download_example.php	– dynamic example-CSV download
├─ lib.php	– navigation hooks
├─ settings.php	– admin settings
├─ db/	
│ └─ access.php	– capability definitions
│ └─ upgrade.php	– config-key migration
├─ classes/	
│ └─ column_config.php	– CSV-header resolver
│ └─ form/upload_form.php	– upload form
│ └─ process.php	– processing engine
│ └─ privacy/provider.php	– privacy API (null provider)
├─ templates/	
│ └─ preview.mustache	– preview table
│ └─ results.mustache	– results overview
├─ lang/{en,de}/	– language packs (English, German)
├─ tests/	– PHPUnit + Behat
├─ docs/user-manual/	– this manual (DE + EN)
├─ README.md	– technical documentation (English)
└─ CHANGELOG.md	– version history

Running the tests

```
# From the Moodle root, after one-time init:  
php admin/tool/phpunit/cli/init.php  
vendor/bin/phpunit public/local/branchupload/tests/process_test.php
```

For the Behat acceptance-test suite see [README.md](#).

Privacy

The plugin is a *null Privacy provider*: it does not store any personal data of its own. All user data flows through the Moodle core APIs (`user_create_user`, `user_update_user`, `delete_user`, `profile_save_data`, `cohort_add_member`), which are themselves Privacy-API compliant.

A full WCAG 2.2 Level AA conformance statement lives in [ACCESSIBILITY.md](#).

Licence and contact

This plugin is released under the **GNU GPL v3 or later**.

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Please send questions, bug reports and feature requests to info@eledia.de, or open an issue on the plugin's GitHub repository.